

1. List all distinct departments

SQL

1 SELECT DISTINCT department

2 FROM Students;

Output

department
IT
HR
Finance

2. Average age per department

SQL

1 SELECT department, AVG (age) AS avg - age

2 FROM students

3 ~~Group~~ GROUP BY department;

Output

department	avg - age
IT	20.5
HR	22
finance	23

3. Departments with more than 1 student

SQL

1 SELECT department, COUNT(*) AS student_count

2 FROM students

3 GROUP BY department

4 HAVING COUNT (*) > 1

Output

Department	Student_Count
IT	2
HR	2

4. Students aged between 21 and 23

1 SELECT *

2 FROM students

3 WHERE age BETWEEN 21 AND 23;

Output

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. IT or HR students older than 21

SQL

1 SELECT *

2 FROM students

3 WHERE department IN ('IT', 'HR')

4 AND age > 21;

Output

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. Total credits per department with more than 5 Credits
SQL

- 1 SELECT department, SUM (credits) As total_credits
- 2 FROM courses
- 3 GROUP BY department
- 4 Having SUM (credits) > 5;

Output

department	total credits
IT	11

7. Courses that do not have 4 Credits

SQL

- 1 SELECT *
- 2 FROM courses
- 3 WHERE credits <> 4;

Output

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Top 3 courses by credits

SQL

- 1 SELECT course_id, course_name, credits
- 2 FROM courses
3. ORDER BY credits DESC
4. LIMIT 3;

Output

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. Maximum, minimum and average grade SQL

1. SELECT

2 MAX (grade) AS max_grade,

3 MIN (grade) AS min_grade,

4 AVG (grade) AS avg_grade

5 FROM enrollments;

Output

max_grade	min_grade	avg_grade
90	78	84.6

10. Number of enrollments per course SQL

1 SELECT

2 Course_id

3 COUNT (*) AS enrolment_count

4 FROM enrollments

5 GROUP BY course_id;

Output

Course_id	enrolment_count
101	1
102	1
103	1
105	1
105	1

11. Total Salary and bonus per department

SQL

1 SELECT

2 department,

3 SUM (salary) AS total_salary,

4 SUM (bonus) AS total_bonus

5 FROM salaries

6 GROUP BY department;

Output

Department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. Departments with average salary above 55,000

SQL

1 SELECT

2 department,

3 AVG (salary) AS avg_salary

4 FROM salaries

5 GROUP BY department

6 HAVING AVG (salary) > 55000;

Output

department	avg_salary
IT	61000
Finance	70000

13. Employees whose salary + bonus is greater than 60,000

SQL

1 SELECT

2 employee id,

3 name,

4 salary,

5 bonus,

6 salary + bonus AS total_compensation

7 FROM salaries

8 WHERE salary + bonus > 60000;

Output

Employee id	name	salary	bonus	total compensation
1	Tom	60000	5000	65000
3	Spice	70000	6000	76000
4	Tyke	62000	5500	67500

14 Total and average budget per department where average budget > 70,000

SQL

1 SELECT

2 department

3 SUM (budget) AS total_budget,

4 AVG (budget) AS avg_budget

5. FROM projects

6 GROUP BY department

7 HAVING AVG (budget) > 70000;

Output

department	total budget	avg_budget
IT	270000	135000
Finance	80000	80000

15. Projects with budgets between 50,000 and 120,000
excluding Marketing

SQL

1 SELECT

2 project_id,

3 project_name,

4 department,

5 budget

6 FROM projects

7 WHERE budget BETWEEN 50000 AND 120000

8 AND department <> 'Marketing';

Output

Project_id	Project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
3	HR portal	HR	50000